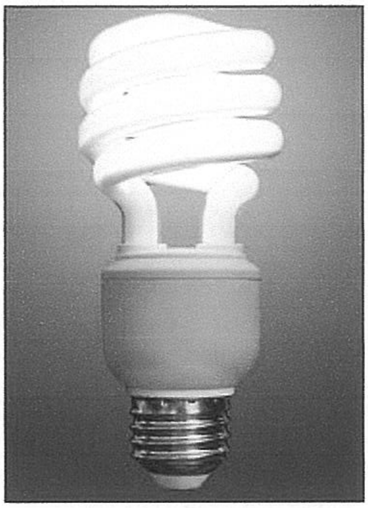
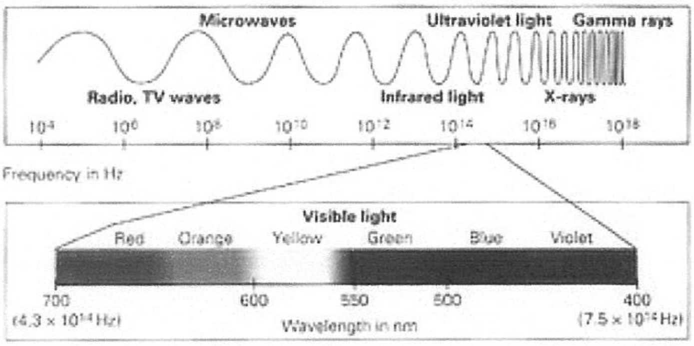


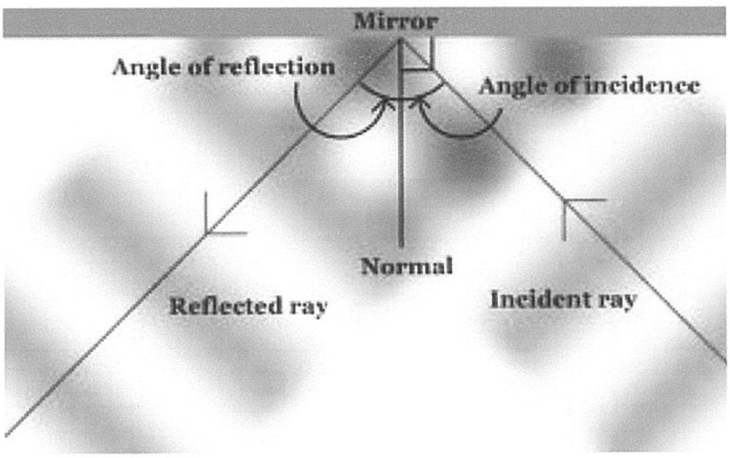


YEAR 9 SCIENCE

PHYSICAL SCIENCE

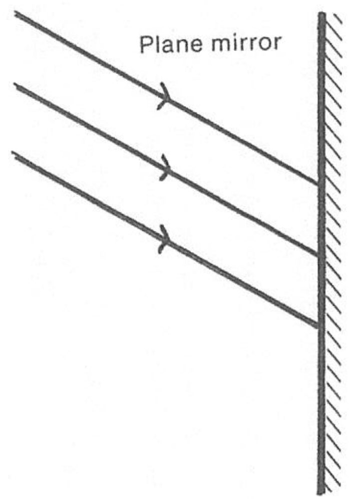
LIGHT

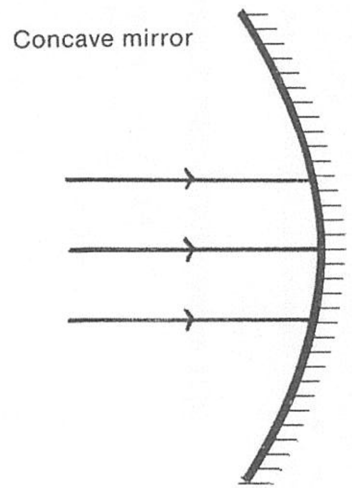
Name: ______________________________

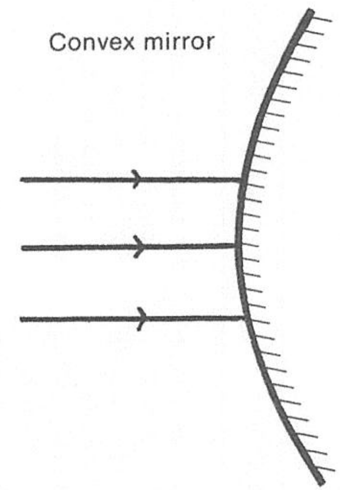


MIRRORS ANGLES REFLECTION

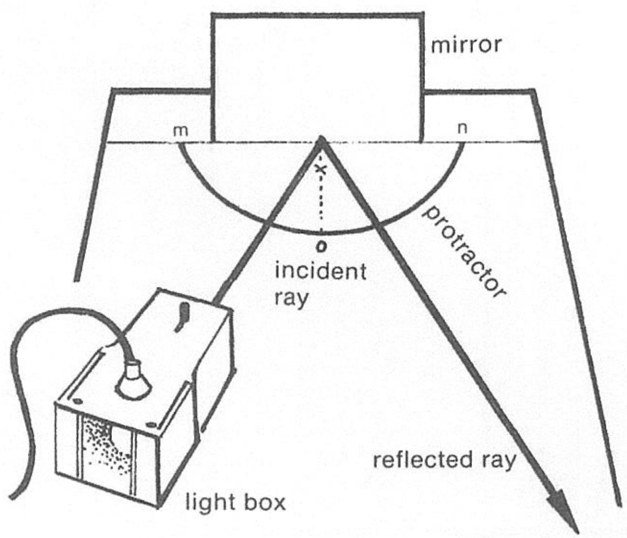
TRACE THE RAY







MEASURE THE ANGLE



PROCEDURE

Using the prepared protractor, place the plane mirror so that its reflecting surface is on the line MN.

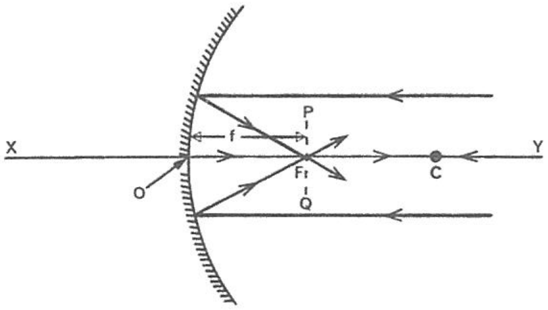
Using a single ray of light, start at O and position the light box at various angle between O and M. The angle between OX and the incident ray is called the incident angle and is recorded in a table of results. The angle of reflection (the angle between OX and the reflected ray) is also recorded.

RESULTS

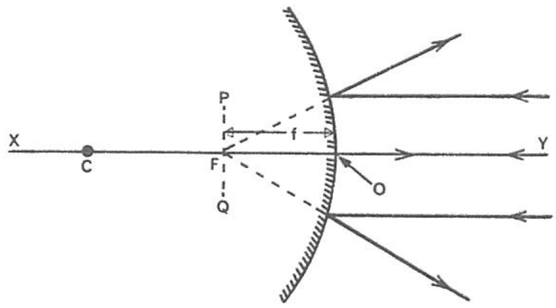
ANGLE OF INCIDENCE	ANGLE OF REFLECTION

CONCLUSION

TERMS USED IN CONNECTION WITH MIRRORS



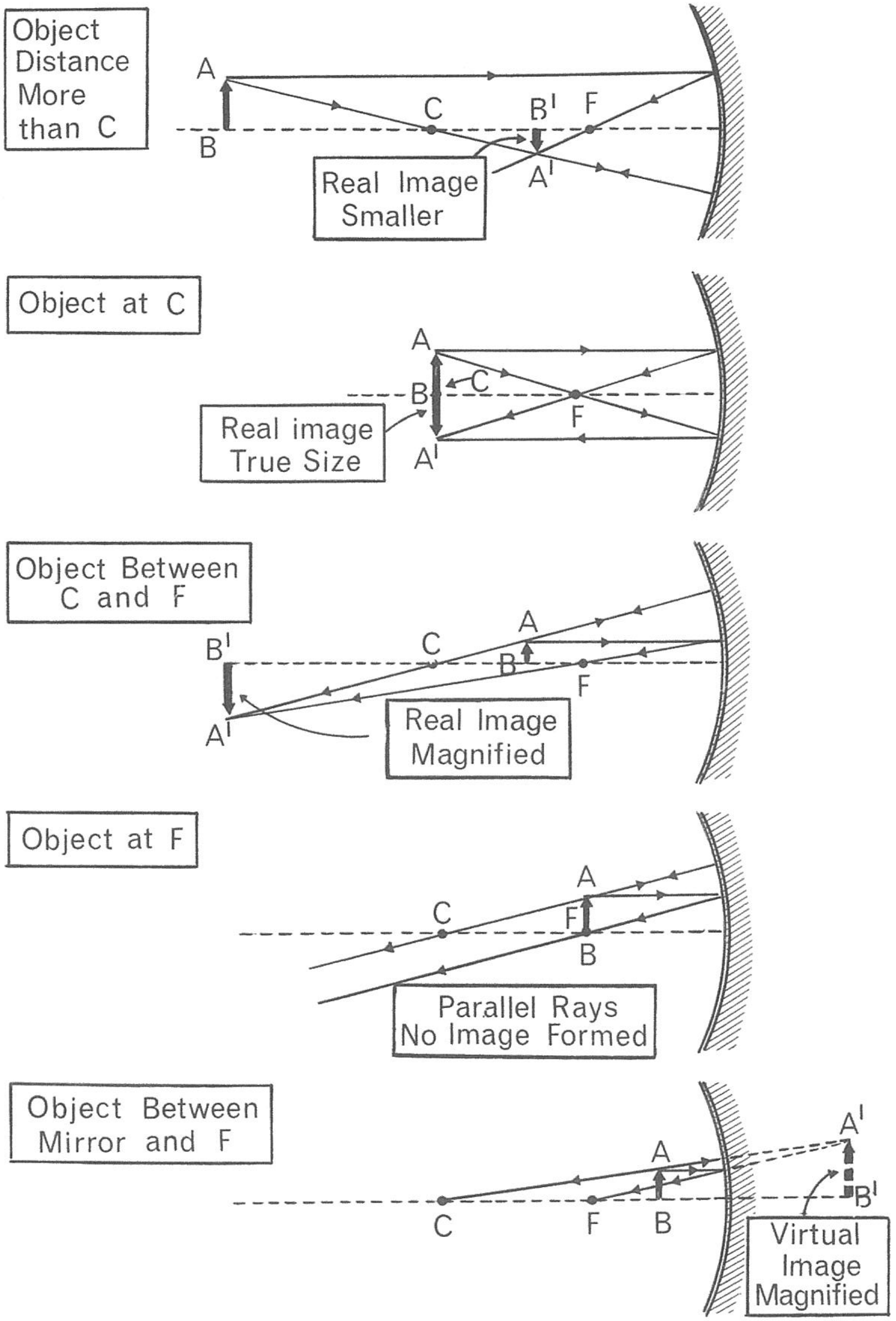
Concave mirror



Convex mirror

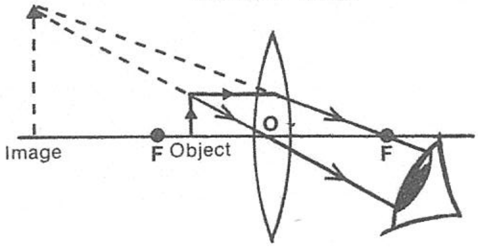
Term		*Description*
Pole	(O)	The centre of the mirror.
Principal axis	(XY)	The straight line passing through the centre of curvature and the pole of the mirror.
Focal point	(F)	The point on the principal axis at which incident rays parallel to the principal axis: (a) intersect after reflection from a concave mirror, or (b) from which parallel incident rays appear to diverge after reflection from, a convex mirror.
Focal length	(f)	The distance between the pole and the focal point.
Radius of curvature	(OC)	The radius of the sphere with centre C of which the mirror forms a part.
Focal plane	(PQ)	The plane that passes through the principal focus perpendicular to the principal axis.

RAY DIAGRAM FOR A CONCAVE MIRROR



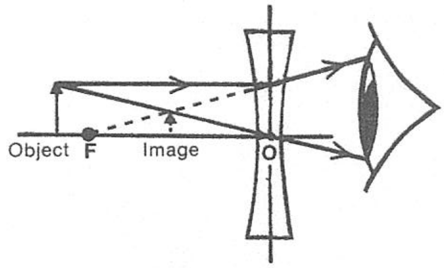
HOW THE EYE SEES VIRTUAL IMAGES

CONVEX LENS



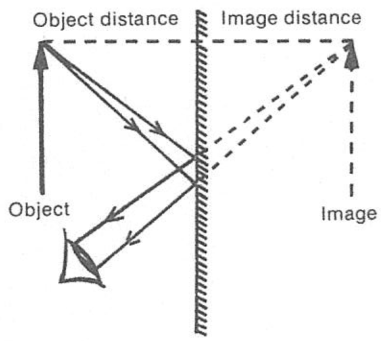
A convex lens will produce a magnified, upright, virtual image when the object is placed between the focal point and the lens.

CONCAVE LENS



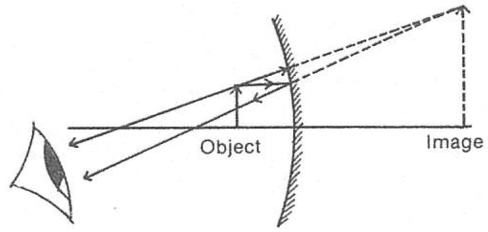
A concave lens always produces an upright, diminished, virtual image.

PLANE MIRROR



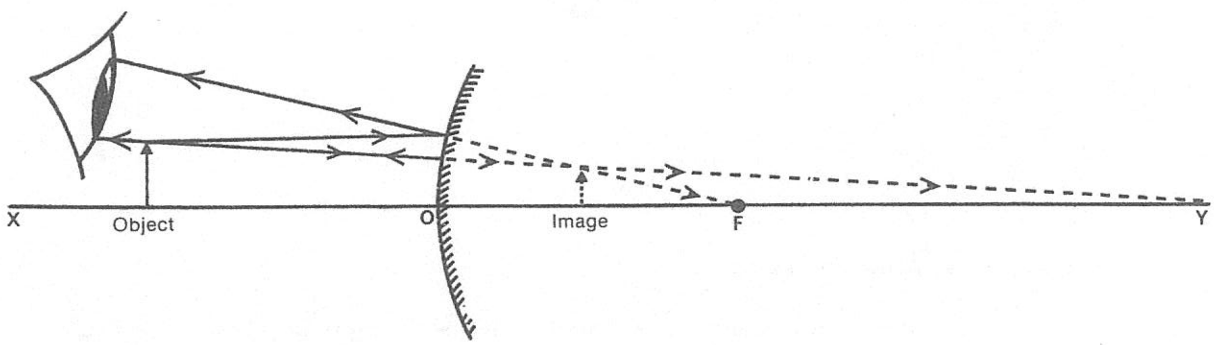
A plane mirror produces an upright, virtual image that is the same size as the object. The object and image distances are equal.

CONCAVE MIRROR



A concave mirror will produce a magnified, upright, virtual image when the object is placed between the focal point and the mirror.

CONVEX MIRROR



A convex mirror always produces a diminished, upright virtual, image.

Images in a concave and in a convex mirror 1a

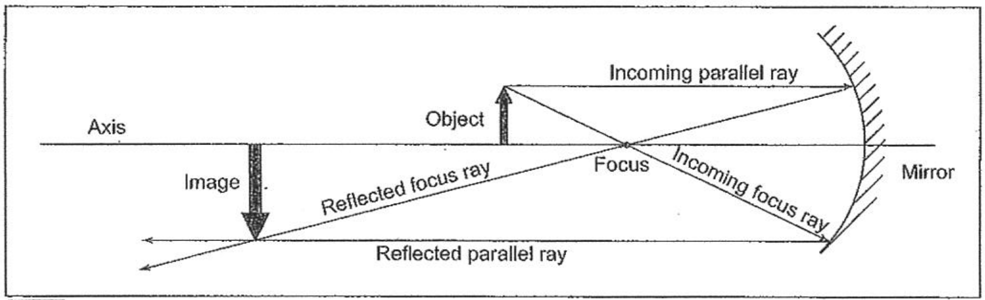
It is possible to predict exactly where the image of an object will appear when that object is in front of a curved mirror. If you select a point on an object (traditionally the tip of a vertical arrow), draw a couple of light rays from that point and follow some rules about reflection, you can find the position of the reflected arrow tip. The position where the two light rays from the object meet or appear to meet after reflection fixes the image.

Here are some things about light rays that you need to know to draw concave mirror diagrams:

- A parallel ray is a ray that is parallel to the axis.
- A focus ray is a ray that travels through the focus.
- An incoming ray travels from the object to the mirror.
- A reflected ray travels away from the mirror.
- After an incoming parallel ray is reflected, it becomes a reflected focus ray.
- After an incoming focus ray is reflected, it becomes a reflected parallel ray.
- If an incoming ray does not hit the mirror, extend it backwards.
- Reflected rays that don't seem to meet might do so if they are extended backwards.
- If reflected rays do not meet, a clear image is not produced.
- If the image is on the same side of the mirror as the object, it is a real image.
- If the image is on the opposite side of the mirror to the object, it is a virtual image.

Compared to its object, an image can be closer to the mirror, further away or the same distance from the mirror. It can be smaller, larger or the same size. It can be upright or inverted, and real or virtual.

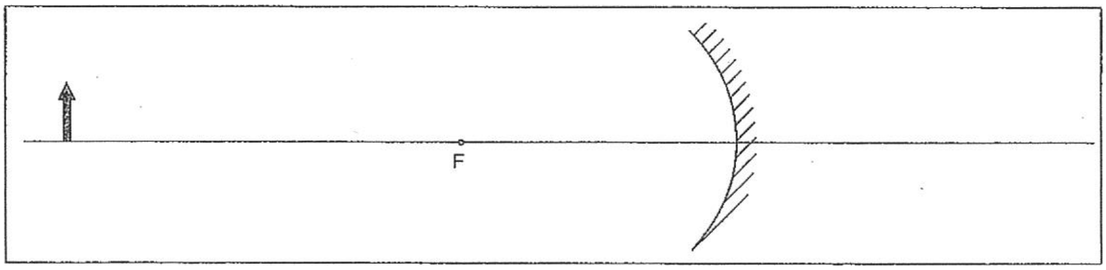
The following diagram shows how drawing a parallel ray and a focus ray from the tip of an arrow can be used to position the reflected tip of the arrow:



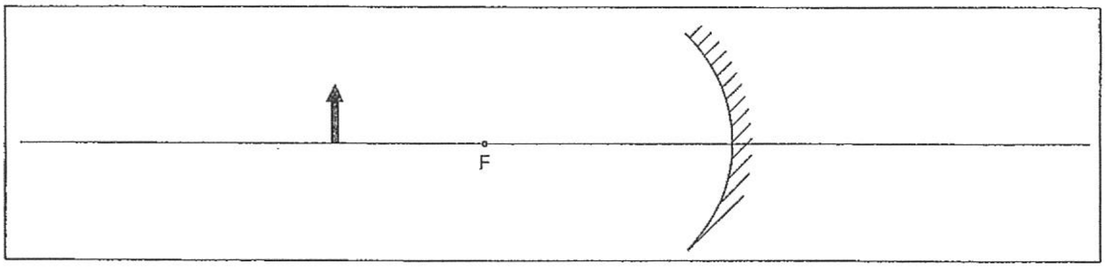
The image is further from the mirror than the object, larger than the object, inverted and real.

1 Construct the image of the object shown in each diagram. Compare image and object.

a

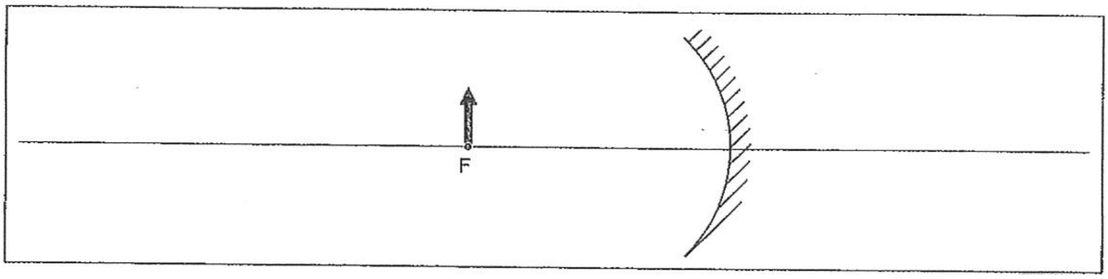


b

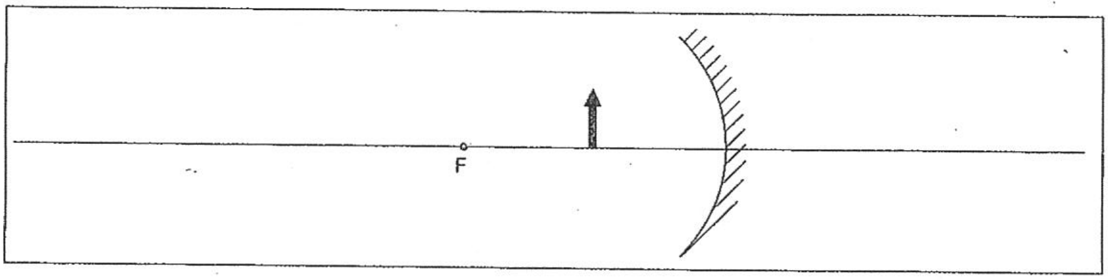


Images in a concave and in a convex mirror

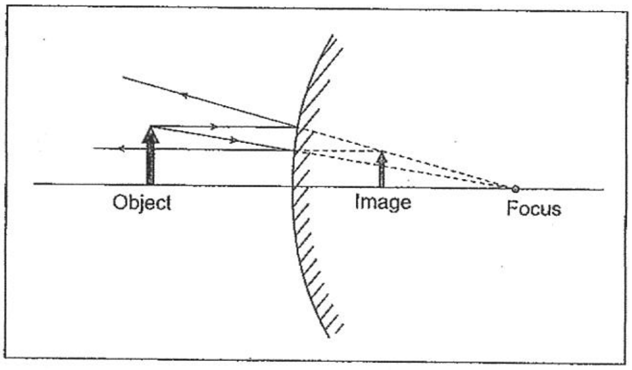
c



d



Below is an object–image diagram for a convex mirror.



The image is closer to the mirror, smaller than the object, upright and virtual.

2 a Develop a set of reflection rules to be followed when drawing convex mirrors.

____

____

____

____

____

b Draw another object–image diagram for a convex mirror in the box at right. Have the object in a different position to that in the diagram above. Apply your rules. Describe the image.

____

____

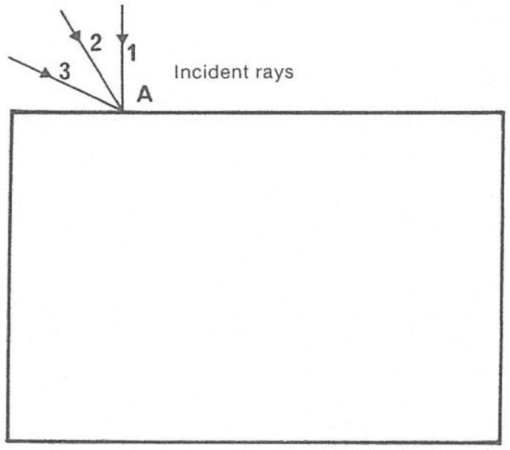
____

____

BENDING LIGHT RAYS

RAYS AND PRISMS

Using the prisms and the light box, shine a single ray on the paths shown on the diagrams. Draw the path of the rest of each ray on the diagrams.



Measure the angle at which ray 1 enters the block.

What do you observe about the path of ray 1?

____

____

Do the incident and emergent rays line up?

____

____

What do you see when you look through the plastic block while the light is shining?

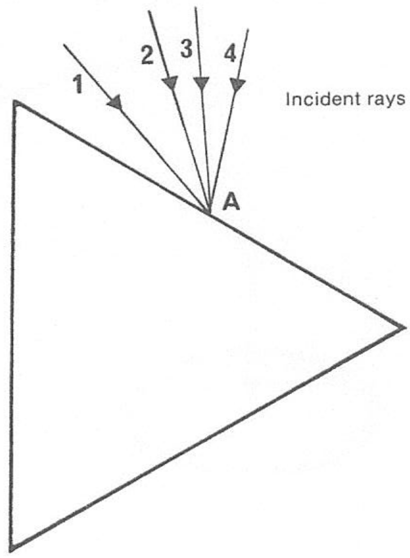
____

____

Do your ray paths look the same as the paths you have drawn?

____

____



Measure the angle at which ray 1 strikes the block.

What do you observe about the path or ray 1?

____

____

Briefly describe what happened to incident rays 3 and 4.

____

____

Do the incident and emergent rays line up?

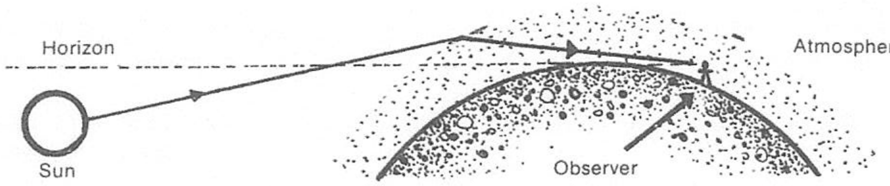
____

____

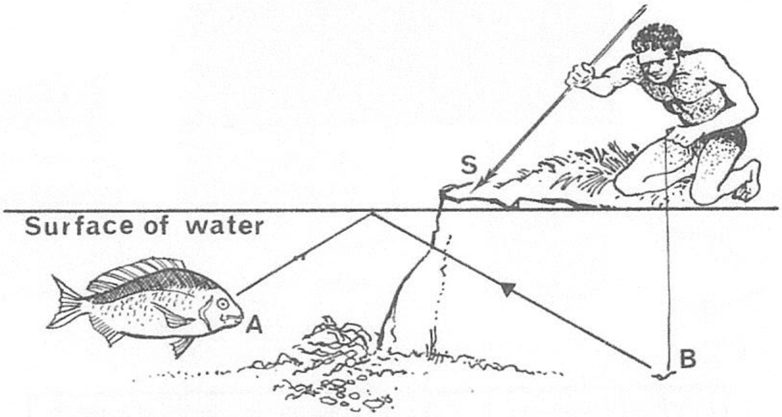
____

8. EXTRAS FOR EXPERTS

Explain why a person is able to see the sun when it is below the horizon. Where is the apparent position of the sun?

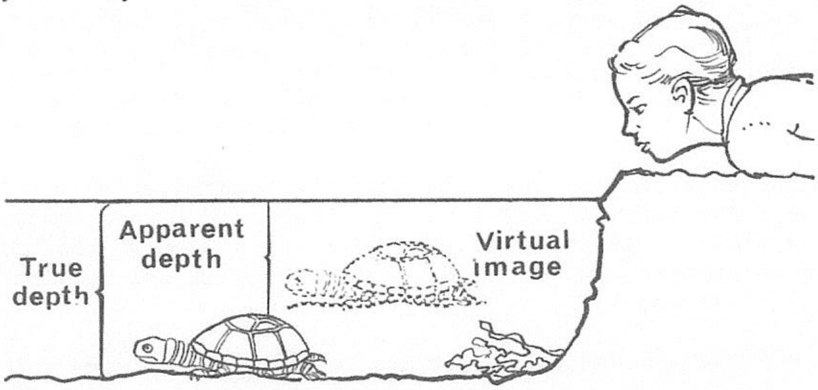


Briefly explain why the fish cannot see the man on the shore, yet the fish is able to see the worm.



Explain why the boat man appears to be rowing with broken oars.

Explain why the turtle is not where it appears to be.



PRISMS AND LENSES

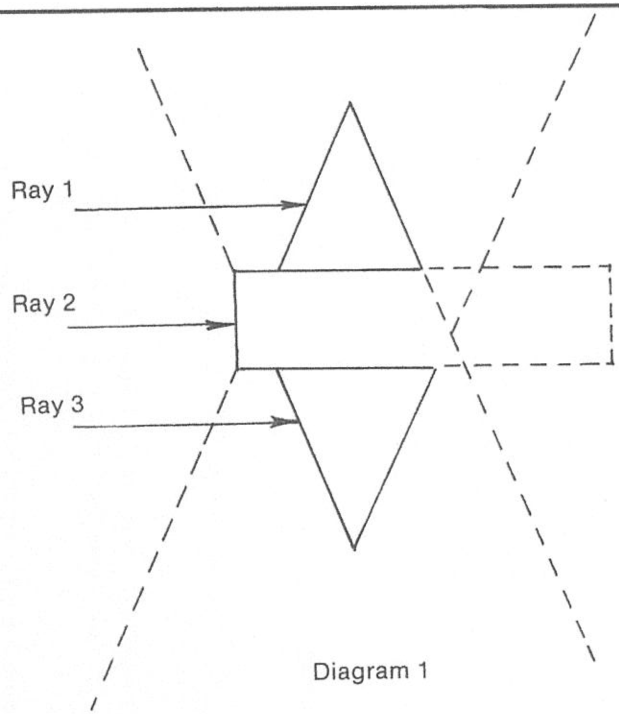


Diagram 1

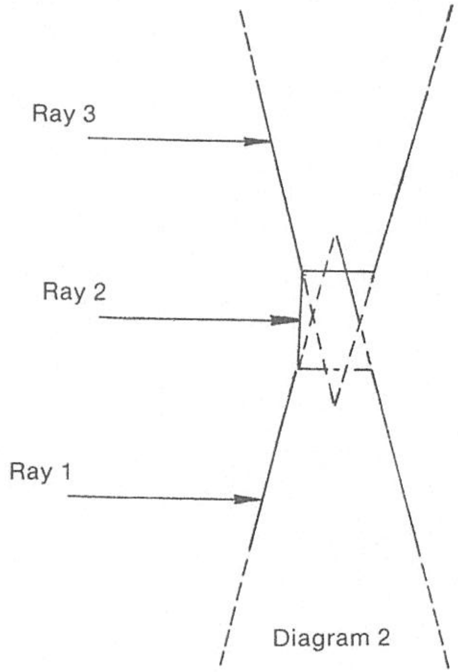


Diagram 2

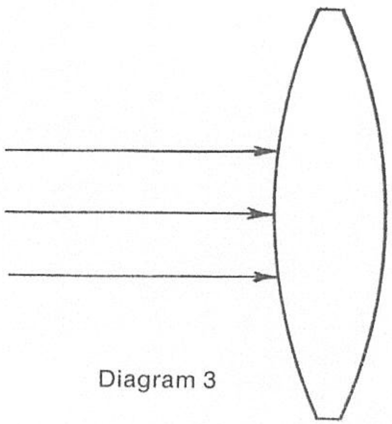


Diagram 3

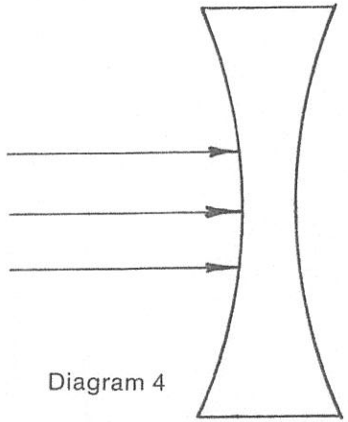
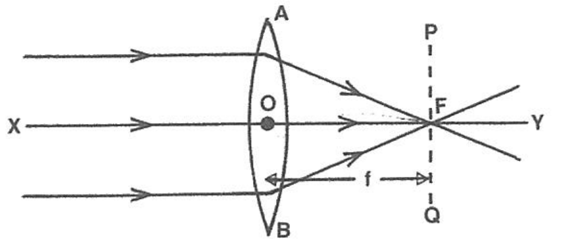
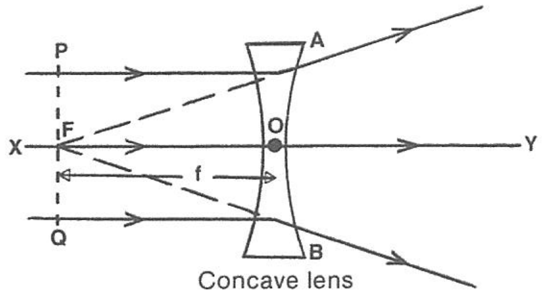


Diagram 4

TERMS USED IN CONNECTION WITH LENSES



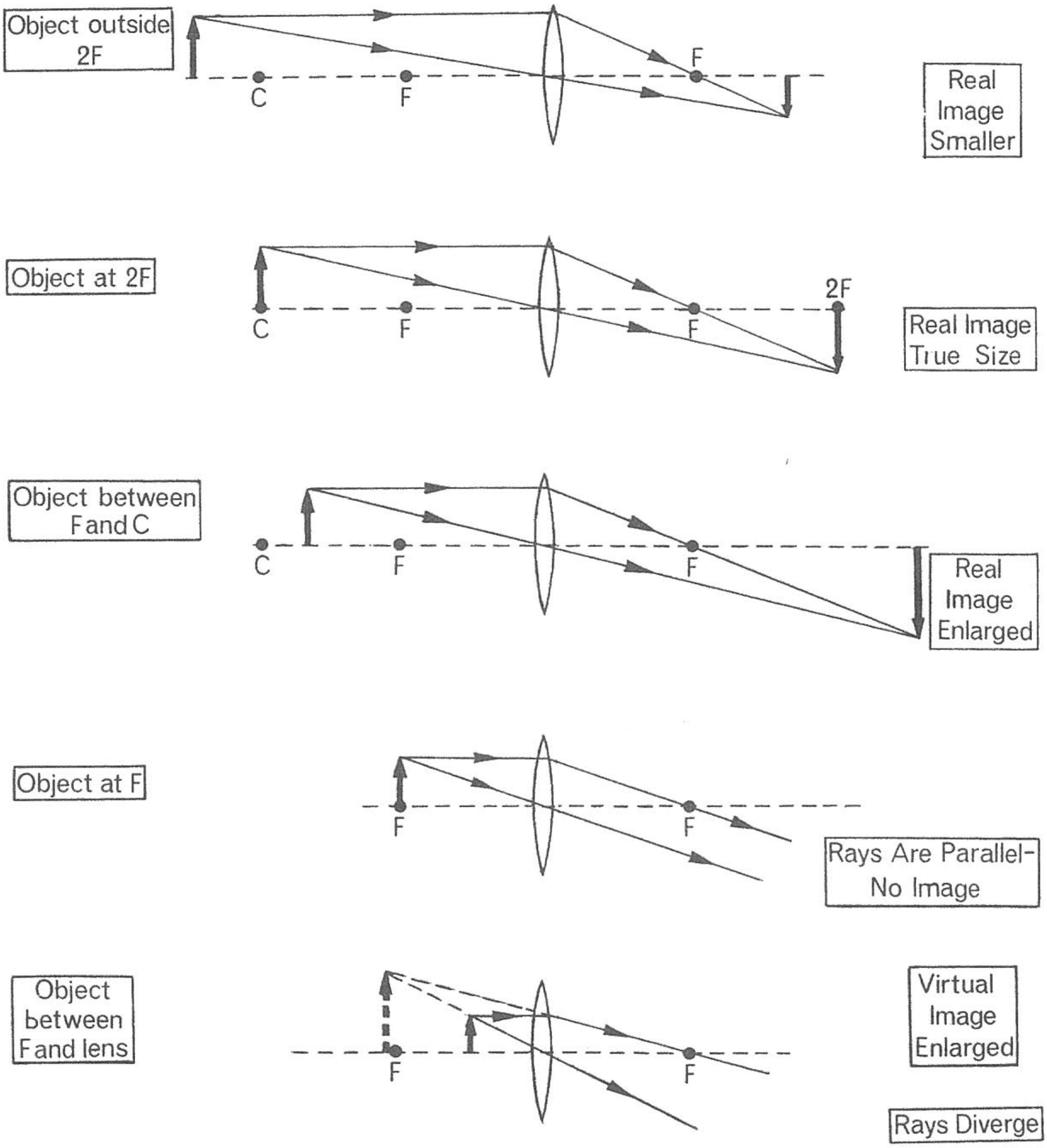
Convex lens



Concave lens

Term		*Description*
Optical centre	(O)	The centre of the lens.
Principal axis	(XY)	The line running through the optical centre at right angles to the plane of the lens.
Focal point	(F)	The point on the principal axis through which incident rays parallel to the principal axis: (a) converge after passing through a convex lens, or (b) from which incident rays appear to diverge after passing through a concave lens.
Focal length	(f)	The distance between the optical centre and the focal point.
Aperture	(AB)	The diameter of the lens.
Focal plane	(PQ)	The plane passing through the focal point and perpendicular to the axis.

RAY DIAGRAMS FOR A CONVEX LENS



Objects and their images in a lens

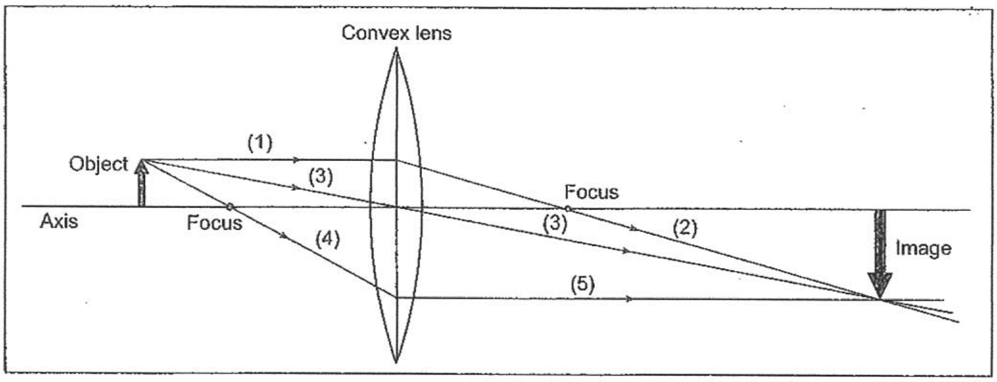
If you want to see an object clearly when using a lens, you have to hold the lens a certain distance away from the object and a certain distance from your eye. It is possible to predict where the focused image will be, if the object is a known distance from the lens. Of course when using a lens, you just move your eye to where the image is clearest and sharpest. To position images by drawing, a couple of rays from a point on an object (e.g. the tip of an arrow) are selected, and drawn following some rules about refraction. The refracted image position of the arrow tip will be given by where these rays meet.

Here are some things about light rays that you need to know to draw images in a convex lens:

- A parallel ray is a ray that is parallel to the axis.
- A focus ray is a ray that travels through the focus.
- A centre ray is a ray that passes through the centre of the lens.
- An incoming ray travels from the object to the lens.
- A refracted ray travels away from the lens.
- Light rays should be drawn as though they refract (i.e. bend) at the middle of the lens.
- After an incoming parallel ray (1 in the diagram below) is refracted, it becomes a refracted focus ray (2).
- An incoming centre ray (3) continues in a straight line through the lens.
- After an incoming focus ray (4) is refracted, it becomes a refracted parallel ray (5).
- If incoming rays do not reach the lens, extend them backwards.
- If refracted rays do not seem to meet, extend them backwards.
- If refracted rays do not meet, a clear image is not produced.
- If the image is on the other side of the lens to the object, it is a real image.
- If the image is on the same side of the lens as the object, it is a virtual image.

Compared to the object, an image can be closer to the lens, further from or the same distance from the lens as the object; smaller than the object, larger or the same size; upright or inverted; and real or virtual.

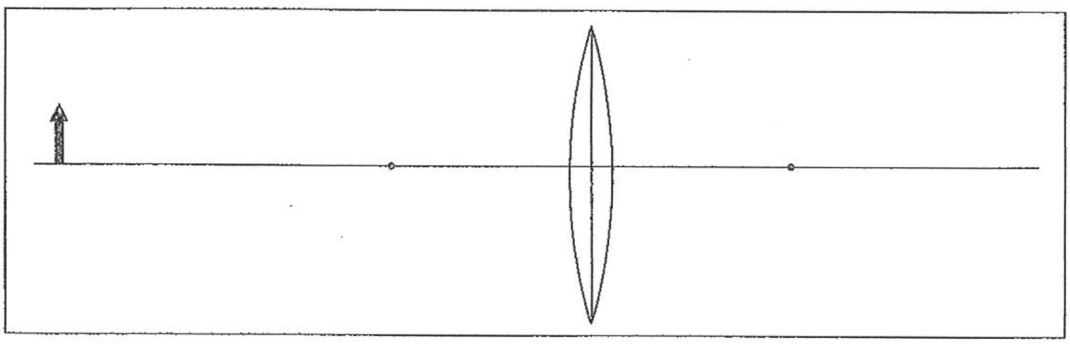
The diagram shows how the image for an object placed near a convex lens can be drawn.



The image is further from the lens, larger than the object, inverted and real.

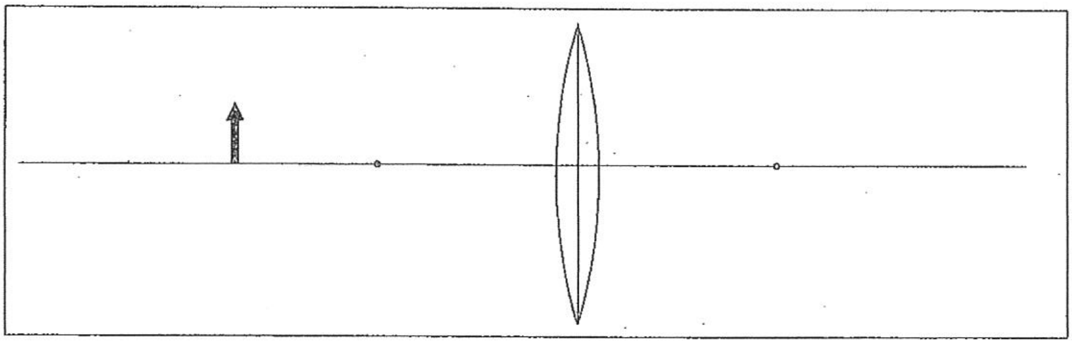
1 Construct the image of the object in each of the following diagrams. Follow the rules listed above. Compare the image with its object.

a

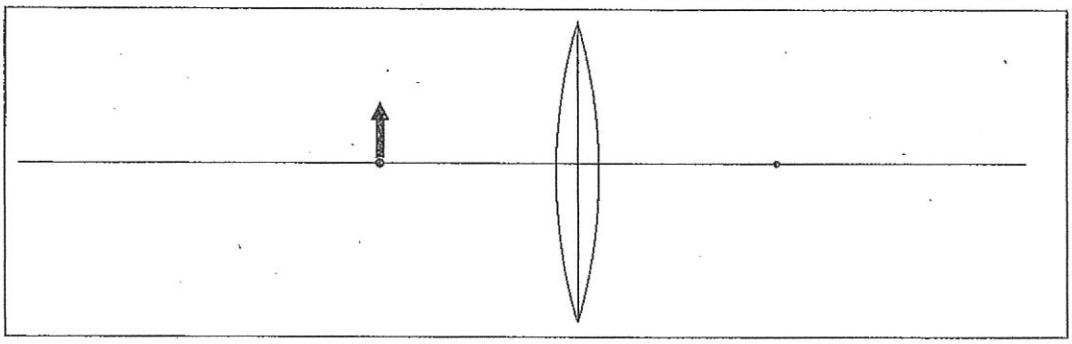


Objects and their images in a lens

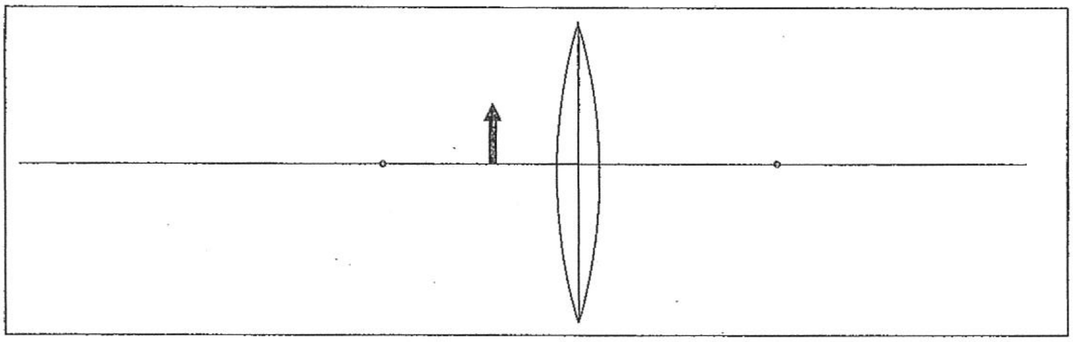
b



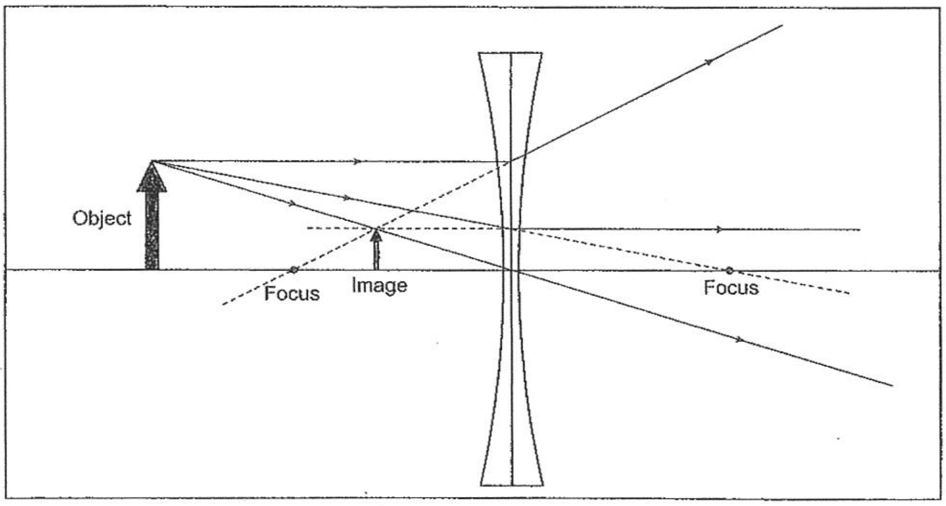
c



d



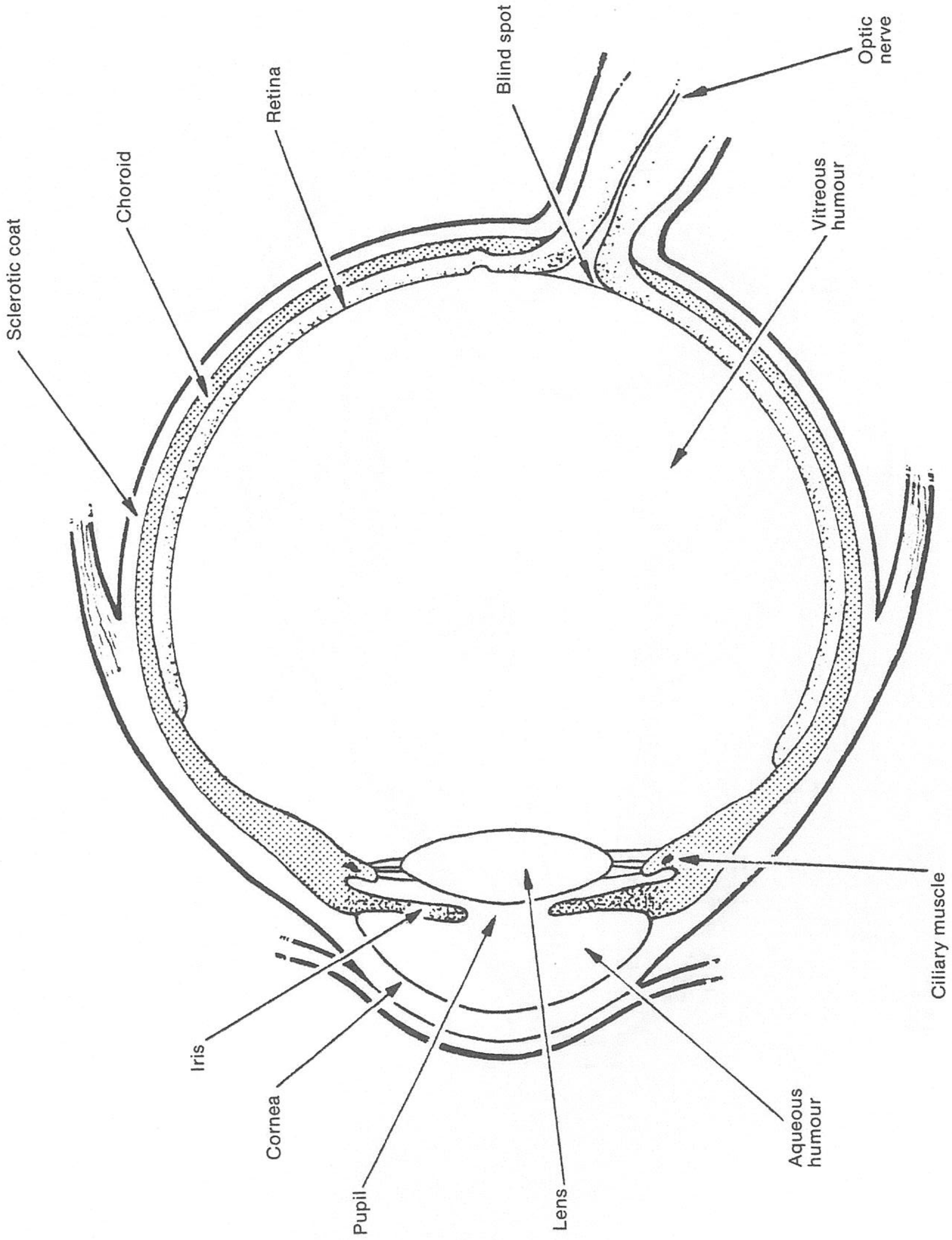
Below is an object–image diagram for a concave lens. The diagram shows the paths of three significant rays that are important in the drawing of the image.



The image is closer to the lens than is the object, smaller than the object, upright and virtual.

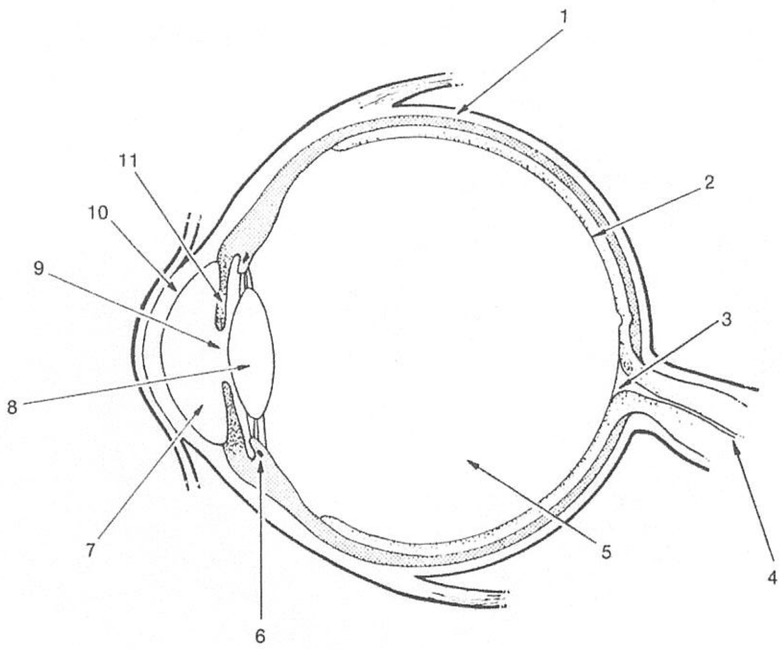
2 Study the diagram and develop a set of refraction rules that will help you draw images produced by concave lenses. Then construct a similar diagram with the object in a different position and apply your rules to find and describe the image. Do these tasks in your science book.

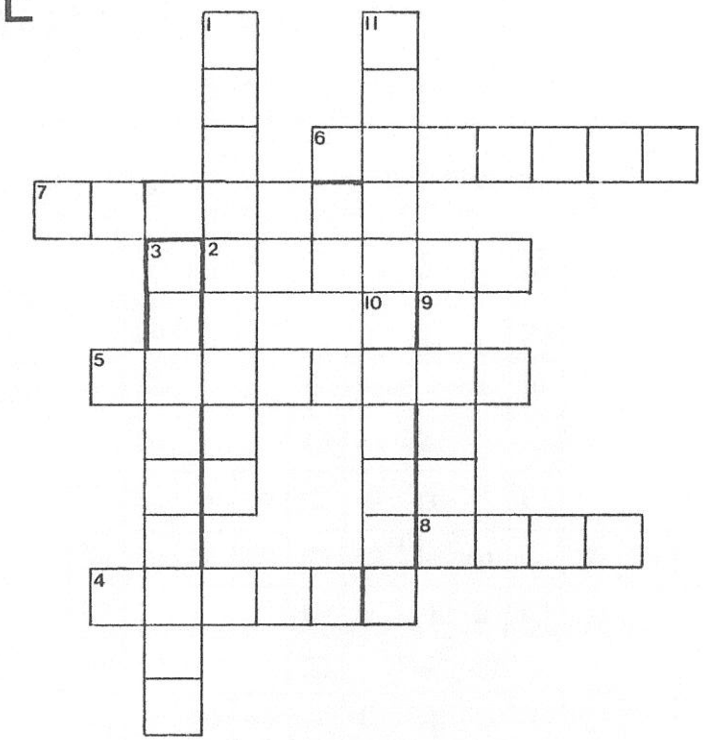
DIAGRAM OF THE EYE



OPTIONAL ACTIVITY

INSIDE THE EYE





The parts of the eye labelled 1 to 11 are the words in the crossword.

ACROSS

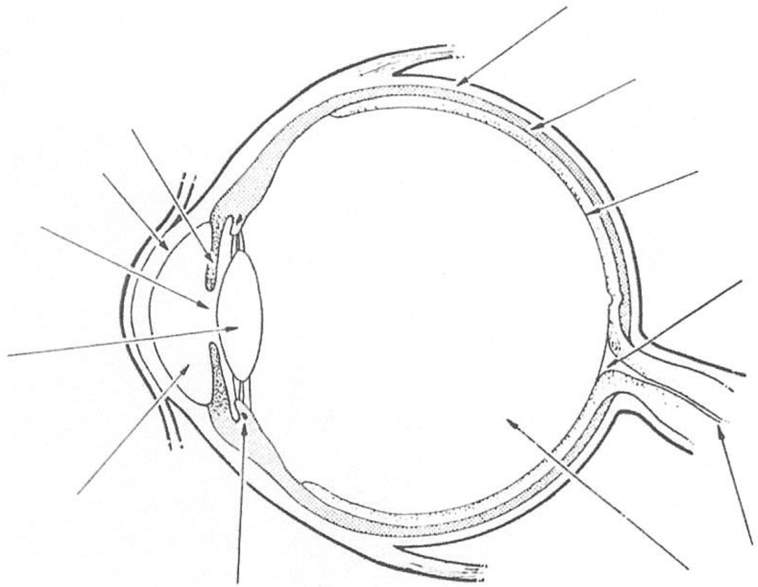
2. An image is formed here.
4. The nerve that carries the message to the brain.
5. A clear, jelly-like substance.
6. A muscle that changes the shape of the lens.
7. A watery fluid.
8. Found in both a camera and the eye.

DOWN

1. Tough coating.
3. No image is seen here.
9. It changes size, depending upon how much light there is.
10. Clear layer at the front of the eye.
11. Different people have different colours.

THE EYE—HOW DOES IT WORK?

LABEL THE PARTS OF THE EYE



All the labels needed to complete the diagram have been listed. Find where each label goes.

Find the function of the following parts of the eye. Briefly describe the function of each.

Aqueous humour, blind spot, choroid coat, ciliary muscle, cornea, iris, lens, optic nerve, pupil, retina, sclerotic coat, vitreous humour.

ANSWER THESE QUESTIONS

1. Explain why it is difficult to see when you go from bright sunlight into a darkened room.

2. Draw diagrams showing the shape of the lens when a person is looking at

   (a) a close object
   (b) a distant object.

3. In what way is a camera's focusing mechanism different from that of the eye?

   Explain how the camera overcomes the problem of focusing on near and distant objects.

4. Describe what an image that is formed on the retina looks like. What must the brain do to the image?

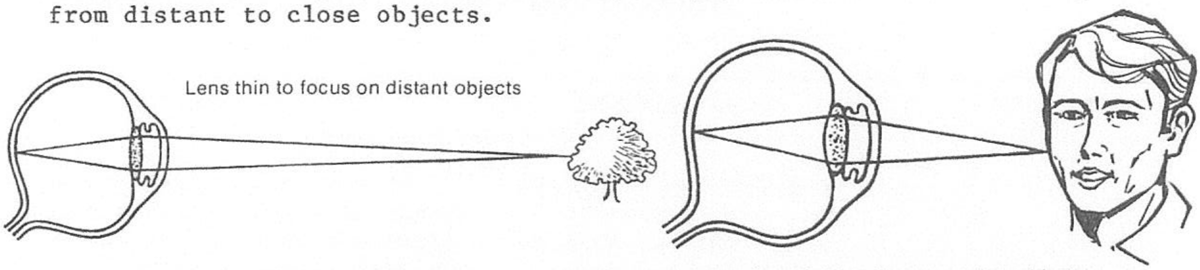
5. Why is it dangerous to look directly at the sun?

6. Cats are said to be able to see in the dark. Find out if this is true. In your answer describe the special features of a cat's eye.

7. Explain what happens when light falls on the blind spot.

DEFECTS OF VISION

ACCOMMODATION. A person with normal eye sight should be able to change focus from distant to close objects.



A camera focuses the image on the film by changing the distance between the lens and the film. A person focuses an image on the retina by changing the focal length of the lens.

Complete the simplified diagrams showing two very common defects of vision.

__________ __________ (MYOPIA)

corrective lens

__________ __________ (HYPEROPIA)

corrective lens

ANSWER THESE QUESTIONS

1. What are bifocal glasses? Describe the type of defect that would require a person to wear bifocal glasses.

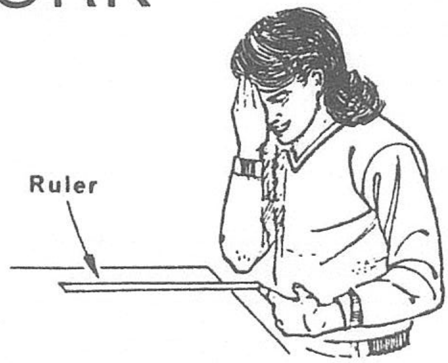
2. Briefly describe what astigmatism is. Can it be corrected?

3. Describe which part of the eye is affected by cataracts. Briefly describe a possible cure for this defect. Explain why you would have to wear glasses after the treatment.

EYES AT WORK

BINOCULAR VISION

Place a ruler so that it hangs approximately 8 cm over the end of the bench.
Standing directly over the ruler, place your hand over one eye and bring the index finger of your other hand from the side and try to touch the ruler. Repeat this several times. Move your head a little closer and try it again.
Record your observations.

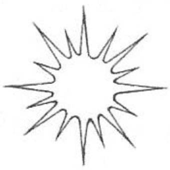


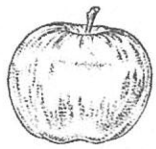
____

____

BLIND SPOT

Hold the page vertically at arms length in front of your eyes. Close your right eye and look at the apple with your left eye. Move the page slowly toward you. What happened?





Record your observations. ______________________________

____

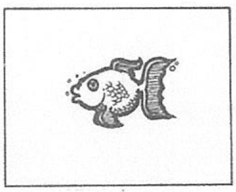
Repeat the procedure, this time close your right eye and look at the star.

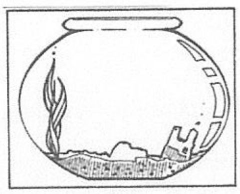
Was the result the same? ______________________________

Explain why this happens. ______________________________

PERSISTENCE OF VISION

On one side of the piece of cardboard draw a fish. Turn the cardboard over and draw a fishbowl. Fit the cardboard into the slot on the top of the piece of dowel.





Spin the card by rolling the dowel between your hands.

Record your observations. __________

____

QUESTIONS

(1) Explain what advantages binocular vision gives people.

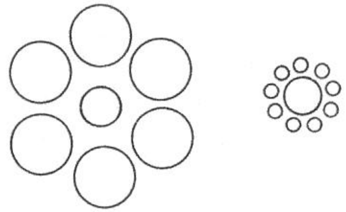
(2) Briefly describe some of the problems a person with one eye would experience.

(3) Explain how persistence of vision is used in movies and television.

SEEING AND BELIEVING

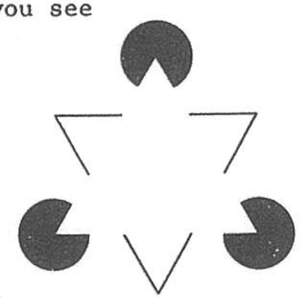
For each of the diagrams below record your observations

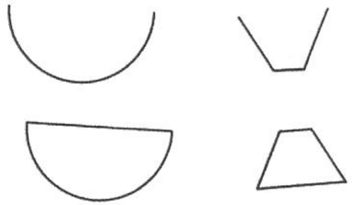
Read each of the signs in turn.
What did you notice?



Compare the sizes of the centre circles. Describe where this type of arrangement could be used.

How many triangles can you see in the diagram?

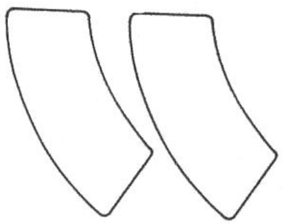




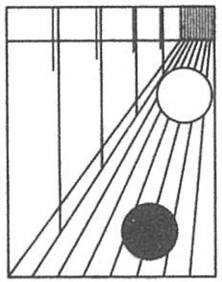
Compare the open and closed figures. How could this influence your judgement of how much a container holds?

Which swimming pool is the larger?

Check your answer by tracing the outline of one pool and placing it over the drawing of the other pool.



Suggest some other examples where this type of outline could be used.



Which of the two shapes is larger?

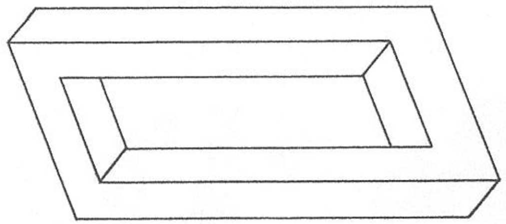
Describe the position of the two circles.

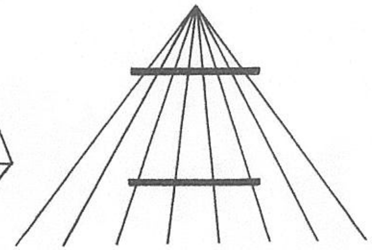
Explain how a photographer uses this effect in reverse.

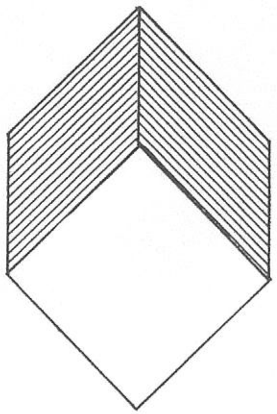
This effect makes large objects appear to move slower than smaller objects travelling at the same speed.

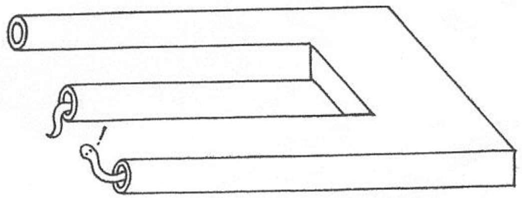
What should this mean to drivers waiting at cross roads and railway crossings?

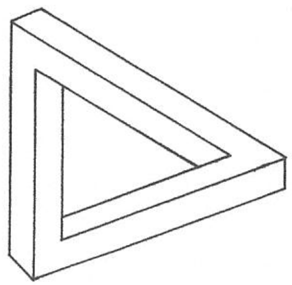
OPTICAL ILLUSIONS

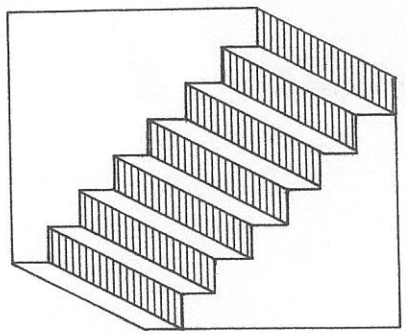






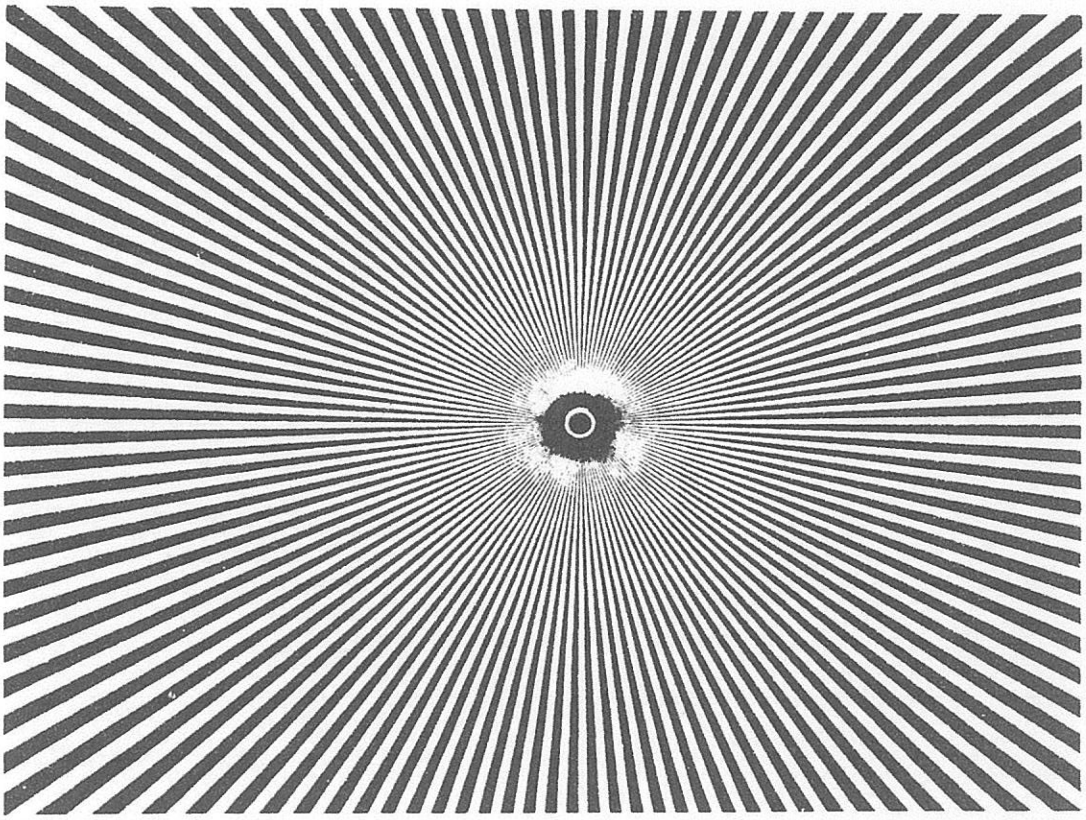






MIRACLE IN THE SNOW

MOVEMENT



COLOUR OF LIGHT

A MYSTERY UNRAVELLED (Complete the quotation when you have finished the activity.)

In the year 1666 ... I procured me a triangular glass prism ... having darkened my chamber and made a small hole in my windowshut, to let in a convenient quantity of the sun's light, I placed my prism at its entrance and saw

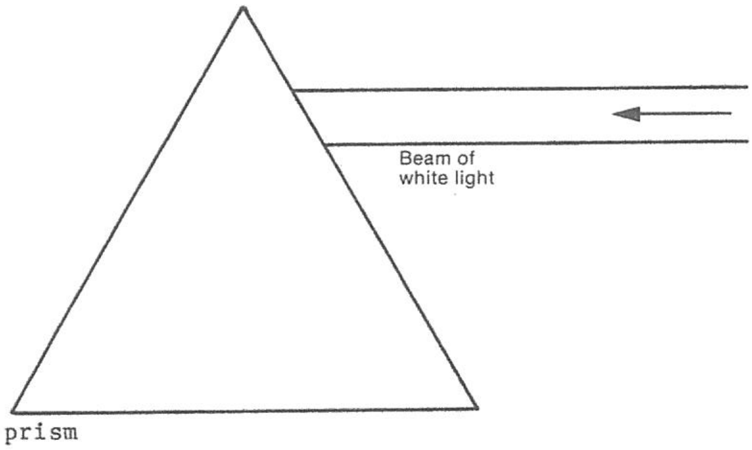
____

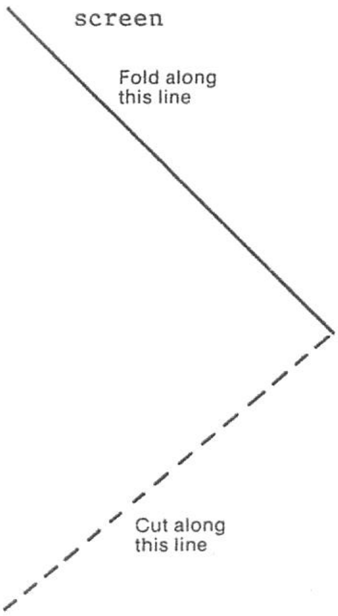
____

____

(*Isaac Newton*)

Set up the equipment as shown in the diagram.





Observation ____

____

____

Answer these questions:

1. What would happen if a ray of red light was beamed down the centre of the beam of white light?

   ____

   ____

2. Explain what could happen if a prism was placed where the screen is. ____

   ____

3. What colour is refracted least by the prism?

   ____

   ____